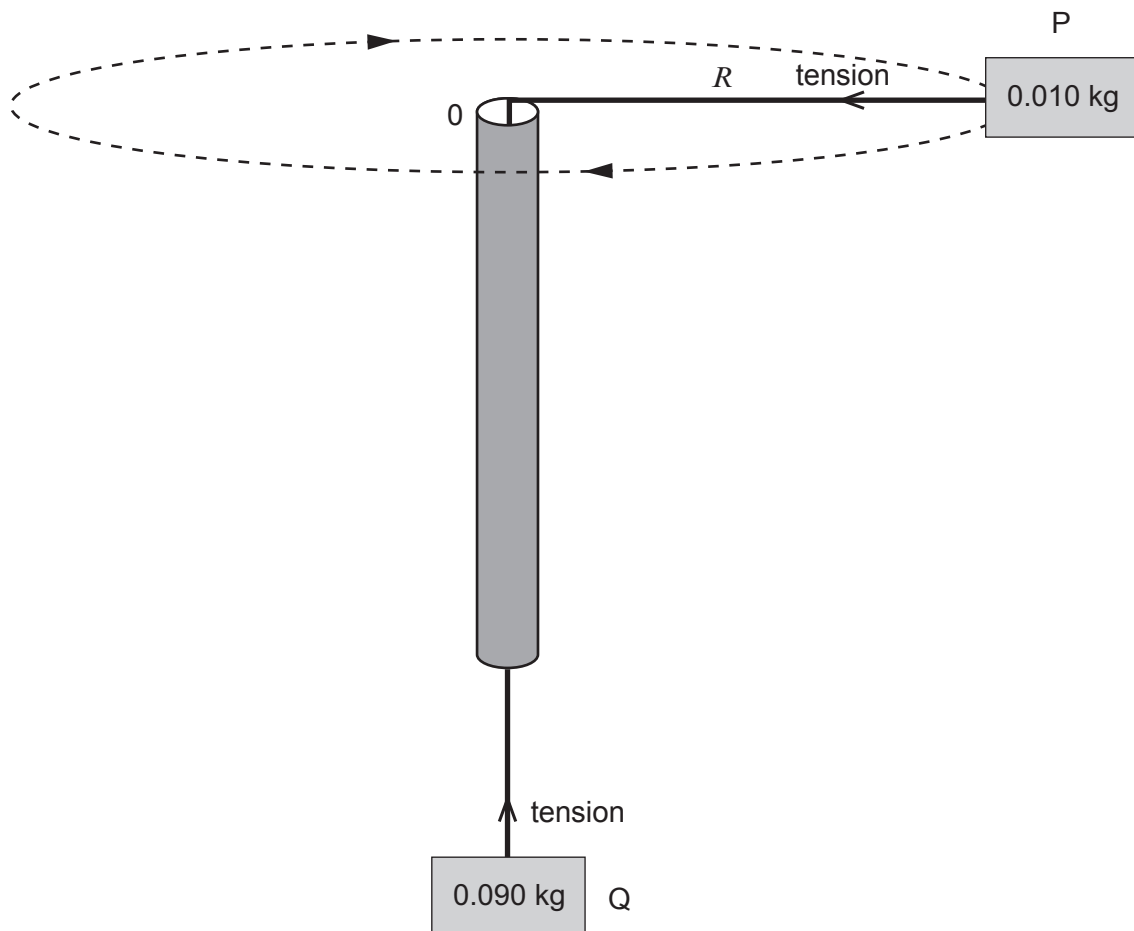


5. A piece of string is threaded through a hollow narrow cylinder. Two small objects, P and Q, with masses 0.010 kg and 0.090 kg respectively are attached to the ends of the string, as shown.

A student holds the cylinder and sets the 0.010 kg mass rotating in a horizontal circle of radius R , which is kept constant at 0.50 m. The time for 10 rotations is recorded. The tension in the string provides both the centripetal force on P and an upward force to hold Q in equilibrium.

The measurement is repeated for different values of R . All measurements are recorded in the table overleaf.



- (a) Show that the speed, v , of mass P for each measurement is given by:

$$v = \frac{2\pi R}{T} \text{ where } T \text{ is the period of rotation.}$$

[1]

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Examiner
only

(b) Complete the table. [4]

R / m	Time for 10 rotations / s	Period T / s	v / ms^{-1}	$v^2 / \text{m}^2 \text{s}^{-2}$
0.50	4.7			
0.60	5.2			
0.70	5.6			
0.80	6.0			
0.90	6.3			

(c) (i) Assuming that OP is horizontal, write an equation relating the centripetal force to v and R . [2]

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(ii) Hence, by using the equation for the forces acting on mass Q, show that:

$$v^2 = 9g R$$

where g is the acceleration due to gravity. [3]

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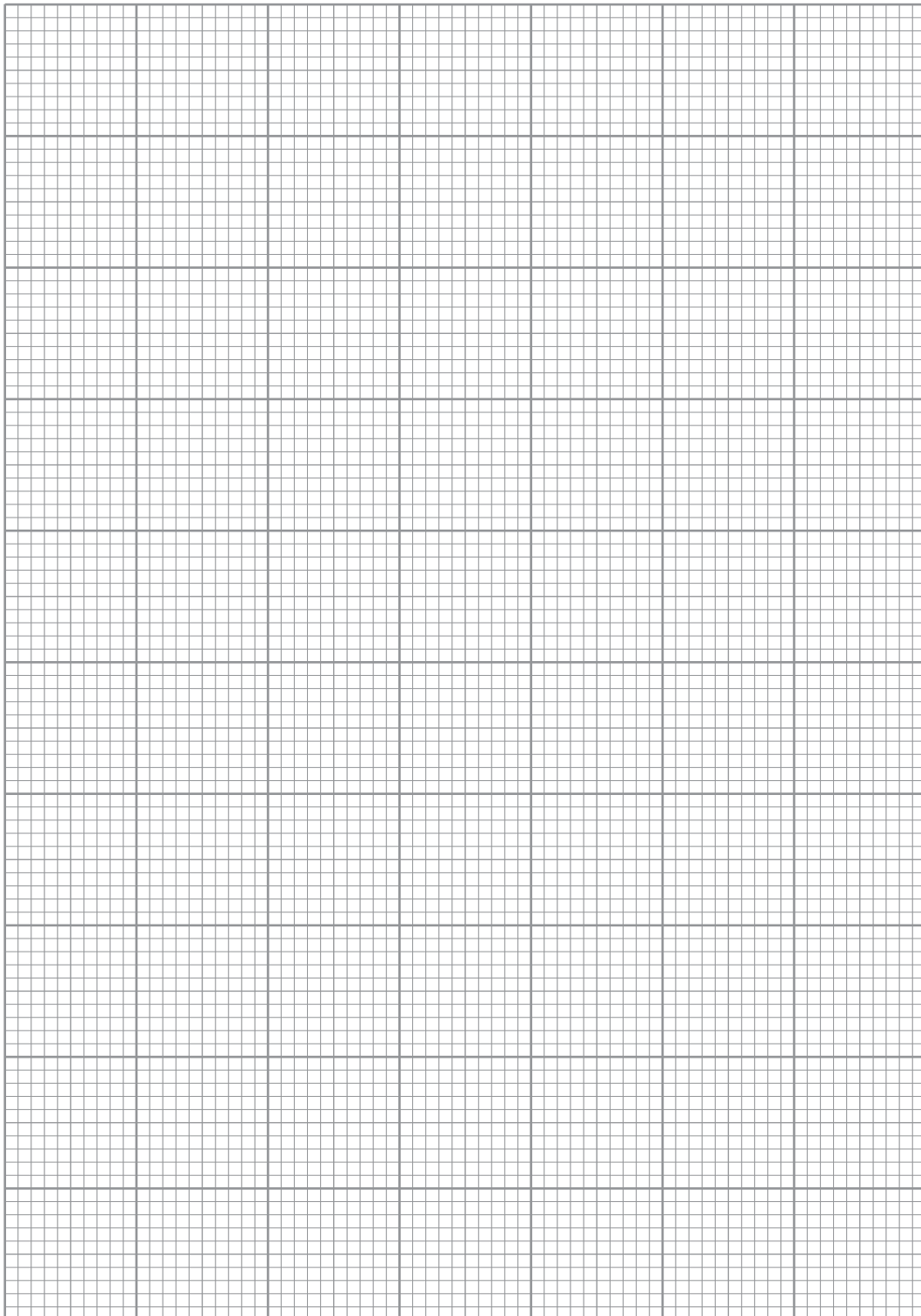
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- (d) (i) Use the data in the table to plot a graph of v^2 (y -axis) against R (x -axis).

[4]

Examiner
only

(ii) Determine a value for g .

[3]

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(iii) Suggest a way in which the experiment can be improved.

[1]

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